Based on OptiCrop Recommendation Engine project :

*Define Problem and Understanding.

1. Specify the Business Problem

Agriculture is one of the most important sectors for food production, but many farmers face challenges in selecting the most suitable crop due to changing weather conditions, soil nutrient imbalance, and lack of technical knowledge. Choosing the wrong crop often results in low productivity, financial losses, and inefficient use of resources. The objective of the OptiCrop Recommendation Engine is to use Artificial Intelligence and Machine Learning to recommend the most suitable crop based on soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH value, and rainfall. This helps farmers make informed decisions and improve agricultural productivity.

2. Business Requirements

Functional Requirements	Non-Functional Requirements
Accept soil and environmental parameters as input.	> Fast response time.
Predict the most suitable crop.	>High prediction accuracy.
Display crop recommendation instantly.	> Easy to use interface.
Provide a simple web interface using Flask.	>Reliable and scalable system.
Store and process crop dataset.	> Secure handling of user inputs.

3. Literature Survey

Several research studies have shown that Machine Learning algorithms significantly improve crop prediction accuracy by analyzing environmental and soil conditions. Algorithms such as Random Forest, Decision Tree, and Support Vector Machine have been widely used in precision agriculture. Recent studies emphasize that AI-driven crop recommendation systems increase productivity, reduce resource wastage, and support sustainable farming. Based on these findings, the OptiCrop Recommendation Engine adopts machine learning techniques to recommend the most suitable crop for farmers.

4. Social and Business Impact

Social Impact, Helps farmers select suitable crops, Increases crop productivity, Reduces agricultural losses, Supports sustainable farming, Improves farmer awareness of AI technology.

Business Impact, Better utilization of land and water, Reduced production costs, Supports smart farming initiatives, Better Farming _ Higher Profit, Encourages digital transformation in agriculture.